UNITED ARAB EMIRATES MINISTRY OF EDUCATION
الإمارات العربية المتحدة وزارة التربية والتعليم

Al Khalidiya Boys School – Cycle 3

School code 10048 | Dibba Al Hisn | Sharjah Eastern

Grade 10 Physics Final Exam Practice

Term 1 2026–2027 • Set A • With Answers

100 marks | 150 minutes | Calculator allowed

Answer all 20 questions. Choose one option for each MCQ. Show working, units and labelled diagrams for Q17–20. Use g = 9.8 m/s² when needed.

Section A Multiple Choice 60 marks

1. Parallel rays strike a rough wall. Which statement explains diffuse reflection? [4 marks]

A. Different local normals send reflected rays in different directions
B. The wall creates new light
C. The law of reflection no longer applies
D. Every reflected ray travels along the normal

Reference: pp. 4–5 | **Answer A**

2. A real object is 0.50 m in front of a plane mirror. Using the real-is-positive convention, X_i is: [4 marks]

A. −1.0 m
B. +0.50 m
C. −0.50 m
D. +1.0 m

Reference: pp. 5–8; p. 8, Check Your Progress 8,10 | **Answer C**

3. An object lies beyond C of a concave mirror. The image is: [4 marks]

A. Between F and C, real, inverted and smaller
B. At F, always upright
C. Beyond C, real and larger
D. Behind the mirror, virtual and larger

Reference: pp. 11–13 | **Answer A**

4. An object is 36.0 cm from a concave mirror of focal length 16.0 cm. Find X_i. [4 marks]

A. +28.8 cm
B. −28.8 cm
C. +11.1 cm
D. +52.0 cm

Reference: p. 16, Practice Problem 13 | **Answer A**

5. A mirror produces magnification m = −0.50. What does this mean? [4 marks]

A. The image is always virtual
B. The image distance must be negative
C. The image is upright and twice as tall
D. The image is inverted and half as tall

Reference: p. 18, Check Your Progress 22,23,26 | **Answer D**

6. What is the speed of light in chloroform with n = 1.51? [4 marks]

A. 3.00×10^8 m/s
B. 4.53×10^8 m/s
C. 1.99×10^8 m/s
D. 1.51×10^8 m/s

Reference: p. 22; p. 25, Check Your Progress 38 | **Answer C**

7. What conditions are required for total internal reflection? [4 marks]

A. Travel from higher to lower n with incidence above the critical angle
B. Travel from lower to higher n at any angle
C. Equal refractive indices on both sides
D. Normal incidence into any material

Reference: pp. 23–25 | **Answer A**

8. Why does a prism separate white light into colours? [2 marks]

A. All colours travel with identical speeds in glass
B. The prism changes each photon's frequency
C. Reflection alone creates new colours
D. Its refractive index depends on wavelength

Reference: pp. 24–25 | **Answer D**

9. To see a rainbow in the late afternoon, you generally look: [4 marks]

A. Directly at the Sun in the west
B. Straight down regardless of the Sun
C. Toward rain in the east with the Sun behind you
D. In the same direction as the Sun

Reference: p. 25, Check Your Progress 42 | **Answer C**

10. A real object viewed through a single concave lens forms an image that is: [4 marks]

A. Always on the far side of the lens
B. Virtual, upright and reduced
C. Real, inverted and enlarged
D. Always the same size

Reference: pp. 26,28; p. 35, Check Your Progress 54 | **Answer B**

11. Which lens defect produces coloured fringes because different colours focus at different positions? [4 marks]

A. Specular reflection
B. Chromatic aberration
C. Total internal reflection
D. Spherical aberration only

Reference: p. 31 | **Answer B**

12. Which lens corrects myopia? [4 marks]

A. Plane glass with no optical power
B. A mirror
C. Convex/converging
D. Concave/diverging

Reference: pp. 32–33; p. 35, Check Your Progress 57 | **Answer D**

13. A bright fringe in a double-slit pattern occurs when the path difference is: [4 marks]

A. Independent of wavelength in every case
B. An integer multiple of wavelength
C. An odd half-integer multiple of wavelength
D. Always zero intensity

Reference: pp. 42–44 | **Answer B**

14. Thin-film colours arise mainly from interference between light reflected from: [4 marks]

A. Only one surface with no second path
B. Two unrelated distant stars
C. The film's front and back surfaces
D. Two opaque screens

Reference: pp. 46–47 | **Answer C**

15. Which object can act as a reflection diffraction grating? [4 marks]

A. A single large opening only
B. A compact disc with closely spaced tracks
C. A flat unmarked sheet of clear glass
D. An ideal point source

Reference: pp. 53–54 | **Answer B**

16. What is the main purpose of a grating spectroscope? [2 marks]

A. Make all wavelengths identical
B. Measure mass directly
C. Increase every photon's energy
D. Separate light into wavelengths for analysis

Reference: pp. 53–55 | **Answer D**

Set A Section B Free Response

17. A fixed incident ray strikes a plane mirror at 52.0° to the normal. The mirror rotates by 35.0° so that the incidence angle decreases. Also consider the convex mirror shown below. [10 marks]

Reference: p. 6, Example Problem 1; pp. 9–14

(a) State the law of reflection and calculate the new angle of reflection. [3]

Answer Angle of incidence equals angle of reflection [1]; new incidence = 52 − 35 = 17° [1]; new reflection = 17° to the new normal [1].

(b) Through what angle does the reflected ray turn? [2]

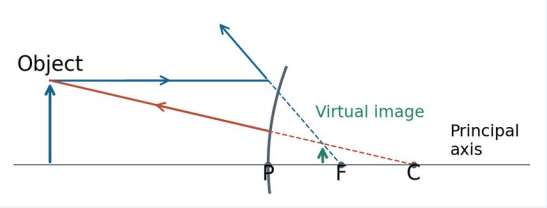
Answer Reflected ray turns through twice the mirror rotation [1], 70.0° [1].

(c) For the convex mirror, define the principal axis and principal focus. [2]

Answer Principal axis: line through pole P and centre of curvature C [1]. Focus F: point behind the mirror from which paraxial parallel reflected rays appear to diverge [1].

(d) Complete a two-ray diagram and state the image properties. [3]

Answer Parallel incident ray reflects as if from F [1]; ray directed toward C retraces, with backward extensions meeting behind mirror [1]; image virtual, upright and diminished [1].



18. A laser passes from air (n = 1.00) into ethanol (n = 1.36) at 37° to the normal. Separately, a 2.25 cm object is 8.5 cm in front of a converging lens of focal length 5.5 cm. Use X_o and X_i for object and image distance. [10 marks]

Reference: p. 21, Practice Problem 30; p. 30, Practice Problem 43

(a) Use Snell's law to find the refracted angle in ethanol. [3]

Answer $n_1 \sin\theta_1 = n_2 \sin\theta_2$ [1]; $\sin\theta_2 = \sin37°/1.36$ [1]; $\theta_2 = 26.3°$ [1].

(b) Calculate the critical angle for ethanol to air. [2]

Answer $\sin\theta_c = 1/1.36$ [1]; $\theta_c = 47.3°$ [1].

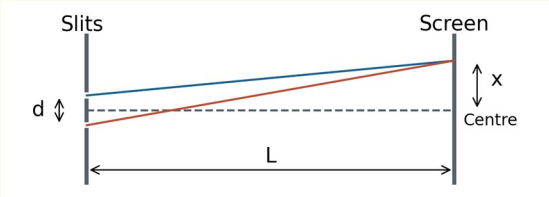
(c) Find the lens image distance. [3]

Answer $1/f = 1/X_o + 1/X_i$ [1]; $X_i = fX_o/(X_o - f) = 5.5(8.5)/3.0$ [1] = +15.6 cm [1].

(d) Find magnification and image height, including orientation. [2]

Answer $m = -X_i/X_o = -1.83$ [1]; $h_i = mh_o = -4.13$ cm, inverted [1].

19. In a double-slit investigation the slit separation is 0.0190 mm, the screen is 0.600 m away, and the first bright band is 13.2 mm from the central band. [10 marks]



Reference: p. 45, Practice Problem 1; pp. 42–44; p. 49, Check Your Progress 13

(a) On the sketch, identify d, L and x, and state the condition for a first-order bright band. [3]

Answer d: slit separation; L: screen distance; x: distance from central band [2 for all three, 1 for two]. Path difference = λ [1].

(b) Calculate the wavelength using the small-angle approximation. [3]

Answer $\lambda = xd/L$ [1]; convert x and d to metres [1]; λ = 418 nm [1].

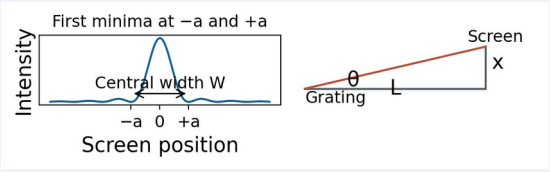
(c) Find the distance of the second-order bright band from the central band. [2]

Answer $x_2 = 2x_1$ [1] = 26.4 mm [1].

(d) Describe the central fringe and nearby fringes if white light replaces the single wavelength. [2]

Answer Central fringe is white [1]; nearby fringes are coloured, with violet closer to centre than red [1].

20. Green light of wavelength 546 nm passes through a single slit of width 0.095 mm onto a screen 0.75 m away. Separately, a DVD grating produces first-order spots at x = 1.29 m from the central spot on a wall L = 1.25 m away using 532 nm light. [10 marks]



Reference: p. 53, Practice Problem 15; p. 56, Example Problem 3

(a) Find the width of the single-slit central bright band. [3]

Answer $W = 2\lambda L/w$ [1]; $W = 2(546 \times 10^{-9})(0.75)/(0.095 \times 10^{-3})$ [1] = 8.62 mm [1].

(b) Explain what happens to this central width if the slit is narrowed. [1]

Answer It increases, since W is inversely proportional to slit width [1].

(c) For the DVD, calculate the first-order angle using exact geometry. [2]

Answer $\theta = \tan^{-1}(x/L)$ [1] = $\tan^{-1}(1.29/1.25) = 45.9°$ [1].

(d) Use $m\lambda = d \sin\theta$ to find groove spacing and explain why x/L cannot replace sin θ here. [4]

Answer $d = \lambda/\sin\theta$ for m = 1 [1]; substitution [1]; d = 741 nm [1]. Angle is large, so small-angle approximation is unsuitable [1].